

PaoloMontesel

Low-level engineer

tooling · compilers · reverse engineering

citizenships

italian 
canadian 

links

babush.me 
github.com/thebabush 

email

cv@babush.me 

languages

italian - native 
english - fluent 
japanese - A2 

location

Toronto 
(open to relocation)

experience

- 2024- **Hex-Rays, Remote** Software Engineer (Contract)
🔗 **Binary analysis platform:** kickstarted a new distributed SaaS platform; developed its multi-algorithm similarity engine (vector similarity, string/callgraph hashing, microcode/IR dataflow hashing).
🔗 **Binary ingestion pipeline:** auto-scaling crawlers, analysis workers and a search engine ingesting open-source and proprietary software weekly to generate signatures for downstream analysis services.
- 2024 **Tsuru Capital, Tokyo** Software Engineer
Options market-maker. Worked on the internal trading UI (Rust + ImGui), scenario pricing, and risk modeling over live market data. 3 months internship converted into a permanent role.
- 2021–2023 **Symmetry Investments, London / Remote** Software Engineer (Contract)
Internal security at a multi-strategy hedge fund: reverse engineering and vendor-risk assessments of third-party software and mobile apps, custom tooling.
- 2020–2021 **rev.ng, Milan** Security Research Engineer
🔗 **Big Match:** search engine that identifies the open-source libraries linked into a compiled binary, using GitHub as the corpus. Solo research project.
demo ↗ blog ↗ bashML ↗ HITB Phuket 2023 talk 🇵🇹
🔗 **LLVM-based s390-to-x86 JIT** for LZLabs Software Defined Mainframe; main developer for ~7 months. Now in production at a major Swiss bank.
🔗 **QEMU Hexagon:** bison-driven pipeline from the Qualcomm ISA spec to TCG generators, validated by differential testing against the reference emulator. Collaboration with Qualcomm.
- 2018- **mhackeroni, DEF CON CTF team** Cyber Media Czar (Volunteer)
Run the team's public presence: social media 📧, video 📺 and content production 📽, sponsor relations, merchandise. On site at DEF CON 2023 for logistics and filming when the team won the Hack-A-Sat finals.
- 2019–2020 **Cefriel, Italy** Mentoring, Master in Security
- 2017–2018 **UC Santa Barbara, USA** Research Assistant, Seclab
- 2016 **Unnamed Startup, Padova** CTO
Two-person startup in collaboration with Pittarosso, a large Italian apparel retailer: deep-learning proof of concept for shoe type, brand, and color recognition from in-store video, taken to executive-level talks. Owned dataset creation, CNN training, and evaluation.
- 2015 **NTT DATA, Tokyo** R&D Intern, Automated Reverse Engineering team
Built COBOL reverse-engineering tools, contributed to the team's Haskell codebase, and built web-based visualizations of legacy systems from source-code metadata. Internship arranged through the Vulcanus in Japan program.

selected projects

otelma personal Fed-rate trading system

pi-memento  self-editing agent memory via nested context transactions

nonna  example-based structural code search; reuse-before-write over MCP and LSP

xrefs  fast binary cross-reference extractor for ELF, Mach-O, PE, and dyld caches

llvm-jutsu  a tongue-in-cheek anti-LLM LLVM pass

Turns integer comparisons into pixel comparisons against procedurally drawn hands, exploiting LLMs' inability to count fingers; statically specialized via Futamura projection. Held off Claude Code and Codex on a keygen challenge for a surprisingly long time.

reflex  lexer recovery from compiled flex tables

Recovers the token grammar of a flex-generated lexer from the DFA tables embedded in a binary. Born from a DEF CON CTF challenge; presented at CODE BLUE 2020.

misc reverse-engineering ask me about it

education

2018–2019 **Cybersecurity PhD dropout** Politecnico di Milano, Italy

2013–2017 **Master** of Computer Engineering, 110L/110 University of Padova, Italy

2010–2013 **Bachelor** of Information Engineering, 108/110 University of Padova, Italy

talks

2023 **Big Match: How I Learned to Stop Reversing and Love the Strings**  HITB Phuket

2020 **Reflex: you give me a parser, I give you a token generator**  CODE BLUE Tokyo

awards

2023 **Hack-A-Sat finals winners** (with mhackeroni), satellite CTF Las Vegas, USA

2018- **DEF CON CTF, 4-times finalist** (with mhackeroni) Las Vegas, USA

2019 **SUTD Critical Infrastructure Security Showdown, 3rd place** Singapore

2016 **Data Science Game, 5th place**  (117 teams, 20-team final) Paris, France

2014–2015 **Vulcanus in Japan:** EU-Japan scholarship and internship program Tokyo, Japan

independent security research

MikroTik RouterOS SMB DoS (CVE-2019-16160)  · Outlook Web App / Exchange 2013 (HTML-sanitization 0-day) · SentinelOne (stored XSS) · Naver LINE 2018 Bug Bounty Hall of Fame  · TIM/Telecom Italia 2018 Responsible Disclosure Hall of Fame  · United Airlines bug bounty · others on request.

technical skills

languages  python,  rust,  java,  C/C++,  SQL,  javascript,  assembly (x86/ARM/MIPS/s390),  haskell,  kotlin

machine learning  similarity search,  information retrieval,  scikit-learn,  xgboost,  pandas,  pytorch

compilers LLVM (IR, JIT), QEMU (tcg, frontends) , JIT and emulator profiling, differential tracing.

binary analysis IDA Pro, function similarity algorithms/services, fuzzing, symbolic execution, dynamic instrumentation, (de)obfuscation.

interests

professional CTFs, compilers, reverse engineering, machine learning, signal processing.

personal piano, synths, guitar, trumpet, singing, music production, 2D animation, video-making, standup comedy, writing, cooking.